

Weihai Shen

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EDUCATION

Stony Brook University **NY, USA**
Ph.D. candidate in Computer Science, GPA: 3.88/4.00 Aug. 2020 - May 2026 (Expected)
Advisor: Dr. [Shuai Mu](#)

Arizona State University **AZ, USA**
Master of Computer Science, GPA: 3.89/4.00 Aug. 2018 - May 2020

Nanjing University of Posts and Telecommunications **Nanjing, China**
Bachelor of Geographic Information System, GPA: 89.3/100, Rank: 1/107 Jul. 2010 - Jun. 2014

PUBLICATION

Mako: Speculative Distributed Transactions with Geo-Replication

[Weihai Shen](#), Yang Cui, Siddhartha Sen, Sebastian Angel, Shuai Mu

USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2025

Introduces Mako, a highly available, high-throughput, and horizontally scalable distributed transactional database system built on decoupled replication, fast networks (e.g., DPDK), and vectorized dependency tracking.

Rolis: a software approach to efficiently replicating multi-core transactions

[Weihai Shen](#), Ansh Khanna, Sebastian Angel, Siddhartha Sen, Shuai Mu

ACM European Conference on Computer Systems (EuroSys), 2022

Introduces Rolis, a new speedy and fault-tolerant multi-core transactional database system that leverages a decoupled per-core replication architecture to significantly reduce contention.

DepFast: Orchestrating Code of Quorum Systems

Xuhao Luo, [Weihai Shen](#), Shuai Mu, Tianyin Xu

USENIX Annual Technical Conference (USENIX ATC), 2022

Introduces a programming framework for developing quorum systems that converts conventional asynchronous-style code into a more expressive synchronous style without hurting performance.

SERVICE

External Review Committee for USENIX ATC 2024

AWARD AND HONOR

- Student Scholarship - First Prize (Top 5%), NJUPT, 2011
- Best Student Model, NJUPT, 2011
- Student Scholarship - First Prize (Top 5%), NJUPT, 2012
- Best Student Award, NJUPT, 2012
- Student Scholarship - Second Prize, NJUPT, 2013
- Excellent Graduation Thesis, NJUPT, 2014

PROFESSIONAL EXPERIENCE

Google - Software Engineer Intern

Network Infrastructure Team

Sunnyvale, USA

May 2024 ~ Aug. 2024

ByteDance (TikTok) - Research Intern

Infrastructure System Lab

Mountain View, USA

May 2021 ~ Aug. 2021

ByteDance - Full-time Software Engineer & Data Engineer

User Growth Team & Data Platform Team

Beijing, China

Jun. 2016 ~ Jun. 2018

Baidu - Full-time Software Engineer

Backend Engineering Team

Beijing, China

Apr. 2015 ~ Jun. 2016

Ganji (acquired by 58.com) - Full-time Software Engineer

Backend Engineering Team

Beijing, China

Jul. 2014 ~ Apr. 2015

TEACHING EXPERIENCE

- Graduate Recitation on CSE: 532 Database Systems, Fall 2025
- Teaching Assistant on CSE 535: Distributed Systems, Fall 2022
- Teaching Assistant on CSE 320: Systems Fundamentals II, Spring 2021
- Teaching Assistant on ISE 305: Database Design and Practice, Fall 2020